

April 9, 2026

Dear Editor,

Please consider our manuscript, “PhysFormer: A Physics-Guided Transformer for Physically Consistent Virtual Power Plant Forecasting,” for publication as a Research Paper in *Electric Power Systems Research*.

The manuscript presents an application-oriented forecasting framework for joint virtual power plant component prediction. PhysFormer combines an explicit physical mapping stream, a physics-guided gating and alignment module, a bounded residual output mechanism, and curriculum-based constraint training. The method is designed to improve physical consistency in load, photovoltaic, and wind forecasts under weather covariates while maintaining competitive predictive accuracy.

Using a real three-year virtual power plant dataset, PhysFormer achieves a boundary violation rate of 0.00% with an MSE of 0.0128 and remains competitive with strong Transformer baselines. These results are relevant to power-system operation with renewable integration and align with the applied scope of *Electric Power Systems Research*.

This manuscript is original, has not been published previously, and is not under consideration elsewhere. All authors have read and approved the submitted version.

Thank you for your consideration.

Sincerely,

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